

Module 3: Perception — Way More Than Five Senses

Learning Objectives

1. Discover that your body has far more than five senses — including **interoception**, your ability to sense what's happening inside your body.
2. Understand that your senses don't just passively receive information — your brain **actively adjusts them** based on predictions.
3. Learn why the same physical stimulus can feel completely different depending on **context and expectation**.

Introduction

In elementary school, you learned about five senses: sight, hearing, touch, taste, smell. Solid list. But here's the thing: it's wildly incomplete. Your body actually has dozens of senses, and some of the most important ones aren't on that list at all.

Can you tell when you're upside down? That's your **vestibular sense** (balance). Can you touch your nose with your eyes closed? That's **proprioception** (knowing where your body parts are in space). Do you know when you're hungry, thirsty, or need to use the bathroom? That's **interoception** (sensing your body's internal state).

Your body is covered — inside and out — with sensors. And here's the coolest part: your brain doesn't just passively listen to these sensors. It actively turns their volume up or down based on what it predicts is important right now. Understanding this gives you a completely new way to think about your body.

Key Concepts

1. Interoception: Your Body's Inner Sense

Interoception is your ability to sense what's happening inside your body. It's how you know you're hungry, thirsty, tired, hot, cold, or need to breathe deeper. It's also connected to

emotions — that "pit in your stomach" feeling when you're anxious or the "warm glow" when you're happy are interoceptive signals.

Here's what makes interoception interesting from an active inference perspective: your brain doesn't just receive these signals. It **predicts** what your body should feel like and then checks the incoming signals against that prediction. When you're about to eat lunch, your brain starts predicting "food is coming" and ramps up hunger signals. Your stomach might growl right at noon even if you ate a huge breakfast — because your brain is predicting it's lunchtime.

People vary a lot in how well they can read their interoceptive signals. Some people are very tuned in to their body and can feel subtle changes. Others might not notice they're dehydrated until they have a headache. Getting better at interoception — paying attention to what your body is telling you — can improve your health, your emotional awareness, and your athletic performance.

2. Your Brain Controls the Volume

Imagine you're at a loud party and someone across the room says your name. Suddenly you hear it clearly, even though there's a wall of noise. That's your brain turning up the volume on relevant signals and turning down the irrelevant ones.

Your brain does this with ALL your senses. When you're playing a sport, your brain sharpens visual tracking and dulls pain sensitivity. When you're trying to fall asleep, your brain turns down visual processing and turns up auditory sensitivity (which is why sounds seem louder at night). When you're studying, your brain filters out background noise — unless someone says something interesting.

This "volume control" isn't random. It's based on **predictions about what matters right now**. Your brain predicts what information will be important and adjusts your senses to maximize that information while minimizing distraction. That's active inference at work in your sensory systems.

3. Context Changes Everything

The same physical sensation can feel completely different depending on what your brain predicts about the situation.

A classic example: researchers gave people the same drink, but told half of them it was an expensive energy drink and the other half it was a cheap generic version. The people who thought they were drinking the expensive version performed better on cognitive tests — not because the drink was different, but because their brain's prediction ("this is going to boost my performance") actually changed how their body responded.

This works with pain too. If a doctor tells you "this injection will hurt a lot," it hurts more than if they say "you might feel a slight pinch." The injection is the same; your brain's prediction changed the experience. That's not "all in your head" in a dismissive sense — it's literally how your nervous system processes information.

Try It!

Interoception Check-In. Set three random alarms during your day. When each alarm goes off, stop and do a 60-second body scan. Rate each of these from 1-10: How hungry am I? How tired? How tense are my muscles? How fast is my heart beating? How am I breathing (shallow or deep)? Write down your answers. At the end of the day, look at the patterns. Did your interoceptive signals change throughout the day? Were you surprised by anything you noticed?

Summary

Your body has far more than five senses — including **interoception**, which tracks your internal state. Your brain doesn't passively receive sensory data; it **actively adjusts** what you perceive based on predictions about what matters. **Context and expectations** can literally change how the same physical stimulus feels. Getting better at reading your body's signals is a real, trainable skill. Next module: **Cognition** — how your brain makes sense of all this sensory data and builds models of your body.